

Individual Addendum Template
QM 2023 Capstone Project - Milestone 4
Individual Contribution Statement
Name: Logan Ledbetter
Team: First Row Team
Date: 5/1/26

1. Personal Contribution to Capstone Milestones

Milestone 1: Data Pipeline (Week 5)

Tasks completed:

- Pulled together the main code base and helped set up the reproducibility pipeline
- Imported and structured all datasets used in the project (housing, hurricane, and cost data)
- Worked with teammates to narrow down and finalize the research question

Hours spent: 6-8 hours

Key deliverable: Organized dataset and finalized research direction

Milestone 2: EDA Dashboard (Week 10)

Tasks completed:

- Improved reproducibility and helped build scripts to generate charts and figures
- Developed visualizations to explore relationships between housing and hurricane data
- Contributed to completing and refining the Data Quality Report

Hours spent: 5-6 hours

Key deliverable: Data visualizations and completed Data Quality Report

Milestone 3: Econometric Models (Week 14)

Tasks completed:

- Built and organized a Jupyter notebook to document analysis steps
- Updated and improved the Data Quality Report with additional checks
- Revised README and supported AI Audit documentation for clarity

Hours spent: 4-5 hours

Key deliverable: Clean notebook, updated documentation, and AI Audit support

Milestone 4: Final Investment Memo (Week 14)

Tasks completed:

- Wrote and contributed to the final investment memo
- Designed and organized presentation slides for clarity and flow

- Helped simplify key findings so they were easy for the audience to understand

Hours spent: 3-4 hours

Key deliverable: Final memo and presentation slides

Total Estimated Contribution

Total hours across all milestones: 25 hours

Percentage of team workload: 25%

Role(s) on team: Pipeline and setup support (M1), Visualization and reproducibility support (M2), Documentation and notebook support (M3), Memo lead (M4)

2. One Defended Methodological Decision

Decision: I supported starting with visual analysis of hurricane activity and housing trends before moving into modeling.

Reasoning: During M2, the visuals clearly showed patterns between hurricane damage and increases in housing prices and rent, which guided our later model decisions. From an economic perspective, this makes sense because damage from hurricanes leads to rebuilding, which increases demand and can raise prices. Using visuals first made our approach more grounded since we relied on actual data patterns instead of assumptions.

Alternative considered: We could have gone straight into building models, but that would have been weaker because we wouldn't have fully understood the relationships in the data first.

3. One Key Limitation of Our Analysis

Limitation: Our model does not account for all possible factors that influence housing prices outside of hurricanes.

Why this matters: Housing markets are affected by multiple variables such as interest rates, population changes, and overall economic conditions. If these shift at the same time as hurricane activity, it becomes difficult to separate their effects from the impact of hurricanes.

Potential mitigation: Future work could add more control variables (such as income levels or mortgage rates) or apply more advanced modeling techniques to better isolate the hurricane effect.

4. AI Audit Notes (If Applicable)

AI Tools Used: GitHub Copilot

Specific AI Use Examples:

Example 1: Jupyter Notebook Setup

Prompt: "Help me organize a Jupyter notebook for this project workflow"

Output: AI suggested a structured notebook layout with sections for data loading, cleaning, visualization, and modeling.

Verification: I checked that each section matched our actual workflow and made sure the code ran correctly with our dataset.

Critique: The structure was helpful, but it was too generic at first, so I had to adjust it to better match our specific variables and project steps.

Example 2: Data Pipeline Script

Prompt: Help me build a reproducible data pipeline for generating figures

Output: AI generated code to organize the steps for cleaning data and producing charts automatically.

Verification: I ran the script and confirmed that it produced the same outputs as our earlier manual work.

Critique: The AI missed some file path and formatting details, so I had to debug and modify the script to make it fully reproducible.

Overall AI Use: I used AI for about 20–30% of my work, mainly for structuring code and improving workflow efficiency. All final analysis decisions and interpretations were done independently.

5. Self-Reflection

What did I do particularly well on this capstone?

I think I did well with making the visuals clear and helping present the information in a way that was easy to follow. I also stayed consistent with keeping documentation updated and organized.

What could I have improved?

I could have managed my time better by starting earlier on documentation and updates instead of finishing them closer to deadlines.

What did I learn from this capstone project?

I learned that cleaning and understanding your data is really important before doing any modeling. I also learned how to collaborate better with a team and explain data findings in a simple and clear way.

6. Attestation

By submitting this individual addendum, I affirm that:

All contributions listed above are accurate and honest

I have not exaggerated my role or minimized teammates' contributions

I understand that this addendum may be used to adjust my individual grade relative to the team grade

I take full responsibility for my work and any errors in the sections I authored

Signature: Logan Ledbetter Date: 5/1/26